Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/18/2016
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Project Title:	Eutectic Solidification in a Ternary Alloy		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
Start Date:	08/15/2014	**End Date:**	08/14/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
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Note this is a follow-on project for Dr. Ralph Napolitano's project,"Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy," which ended in 2014. Solidification processing of metallic alloys can produce a wide array of growth morphologies, which can be manipulated to access new structures, materials, and functionalities. The array of accessible structures is particularly wide in ternary and other higher-order systems, but the challenge of controlling the many variables that influence growth structures is immense and requires a better understanding of the interplay between the mechanisms of pattern formation and the influence of applied forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal

Task Description:	field. The two principal objectives to be addressed are (i) to establish an analytical framework for 2D and 3D quantitative description of multiphase microstructures, and (ii) to employ the newly developed quantitative framework to investigate the dynamics of pattern formation in ternary eutectic alloys. The work will focus primarily on the invariant three-phase eutectic structure. Ground-based samples will be processed via a vertical Bridgman-style directional solidification furnace. Electron backscatter diffraction will be a major characterization tool to measure crystallographic orientation, texture, and misorientation relationships. Analysis over the length of the sample will provide insight into initial transient behavior, grain competition, and steady state growth. The relationship between geometric and crystallographic parameters, such as curvatures, orientations, and periodic ordering, will be used to quantify the morphological trajectory of the eutectic and identify fundamental principles of pattern formation. Seeded directional solidification (DS) experiments will be conducted to compare the dynamics associated with specific crystal orientations. In order to validate the experimental correlations and analytical models developed from the findings of the ground-based work, carefully selected flight-based directional solidification experiments will be conducted using the European Space Agency (ESA) Materials Science Lab (MSL) and the Low Gradient Furnace (LGF). The project will continue to provide collaborative support to the European Space Agency's project titled "Solidification along a Eutectic Path in Ternary Alloys" (SETA), which has already accomplished important work using the Al-Ag-Cu alloy system. Significant findings of the SETA project to date have been in the areas of pattern development and the role of fault structure for both univariant and invariant eutectic compositions, the role of crystal orientation and "locked" grains, and development of nonplanar growth modes. Facilitating open scientific interaction between the Principal Investigator (PI) and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. The planning and performance of flight-based solidification experiments is being done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As mentioned in the Task Description, the objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is far-reaching, given the extremely broad range of multiphase structures that can be fabricated through solidification processing, and is certainly not limited to metals and alloys. Indeed, the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
Task Progress:	The technical progress achieved during the second year of this program (August 15, 2015 through August 14, 2016) is outlined here. Specific areas of focus have been (i) continued development and testing of a directional solidification furnace, and (ii) characterization and analysis of the Al-Ag-Cu three-phase invariant eutectic structure after directional solidification, particularly the response of the structure to a change in velocity. During year 1, in order to perform controlled, ground-based experiments at University of Alabama at Birmingham (UAB), the PI and four students worked together to design and build the Bridgman-type furnace. A thermal gradient is created inside the furnace chamber by a Kanthal wire resistance heating element supported by a mullite shell and a water-cooled copper chill block. The heating element was selected specifically to support the range of temperatures needed to investigate Al-Ag-Cu (700-1200°C). The chamber is insulted with two types of zirconia insulation. A screw drive and stepper motor control vertical translation of the furnace unit. Samples will be contained in a suspended crucible. An inert argon atmosphere is maintained by direct attachment of a tube to the crucible. Construction of the furnace was finished in June 2015. During year 2, procedures for controlling both the movement and the temperature of the furnace were developed, and the furnace was used to directionally solidify several samples. Work to quantify the thermal gradient at the solid/liquid interface for different temperature settings is still underway. A major focus during year 2 has been to study the behavior of ternary eutectic patterns in response to a change in solidification velocity. By perturbing an established pattern in this way, we are able to study how the system reacts to forced changes and in doing so, better identify the mechanisms that control pattern formation in this type of complex, multi-phase material. For analysis, a sample was selected that had been directionally solidified at a velocity of 0.7 μm/s for 85 mm, then the velocity quickly increased to 4.0 μm/s for an additional 45 mm. Cross sections were analyzed every 5-10 mm over that range of distances. It was discovered that, as expected, the scale of the microstructure decreased with increasing velocity. However, it was also found that the microstructural pattern changed significantly. At the initial lower velocity, the pattern was relatively well aligned in the so-called 'brick-like' structure. However, after the velocity increase, the pattern becomes initially less ordered, then slowly resumes an increased amount of order, but with a different phase arrangement, where the white Ag2Al phase becomes elongated and the quasi-lamellar regions of alternating intermetallic phase (Ag2Al and Al2Cu) run approximately 90° from the initial lamellar alignment. Electron backscatter diffraction of similar images in the sample confirms that the crystallographic alignment and orientation relationships between each of the three phases stay exactly the same. Therefore, the pattern change must be due to some other factor. It is hypothesized that the samples move from an interface-controlled pattern, that minimized interfacial energies, at low velocities, to a diffusion-controlled pattern, that minimized diffusion distances, at sufficiently high velocities. In order to quantify the changes to these complex structures, significant time was spent developing a method for image processing and procedure for quantification of the microstructure. In order to perform quantitative analysis, the microscopy images must be thresholded. A variety of quantification methods were tested; those selected as most useful were average phase area, phase fraction, spacing, and the shape factor 'circularity'. These image processing and analysis techniques will be important to all future images, and work is ongoing to fully automate the procedure using MatLab. In order to better understand the behavior of the eutectic structure during a velocity change, the PI worked with the research group of Prof. Britta Nestler at the Karlsruhe Institute of Technology in Karlsruhe, Germany, which performs large-scale phase field modeling of 3D ternary eutectic growth in Al-Ag-Cu. Since this collaboration began in year 1, two papers have been published and a third, related to the velocity change question, is in preparation. The simulations all

	of close examination of the specific branching mechanisms used to decrease the scale of the structure, which is much less straightforward than in a binary eutectic. Phase field simulations confirmed a change in volume fraction with velocity, which appeared to be stable, rather than a fluctuation localized after the velocity jump region. Simulations showed that an increase in velocity resulted in a decrease in Al, but an increase in Ag2Al and Al2Cu. Phase fraction adjustments occurred relatively rapidly, while the adjustments to spacing and circularity of the pattern occurred more slowly, i.e., over longer distances. In addition, the effects of a velocity jump were also compared to the results seen by Anne Dennstedt, SETA team member at DLR (German space agency) in Cologne. After directional solidification processing of a ternary eutectic sample with a velocity change from 0.32 to 0.08 μm/s, a small region of the sample was subjected to x-ray tomography at the European Synchrotron Radiation Facility in Grenoble. This sample also showed a change in phase fraction, consistent with the simulation except that the amount of Ag2Al increased slightly in the DLR sample with decreasing velocities. This is attributed to solid-state changes in the experimental samples, as the solubility decrease of Ag in Al causes precipitation of the Ag atoms onto existing Ag-rich phase. This effect is not currently accounted for in the simulation. Another important finding was that the phase pattern in both the DLR sample and the simulation did not change as a result of the velocity jump. It is hypothesized that the change in velocity produces a change in undercooling of the solid/liquid interface, thereby shifting the equilibrium compositions of each phase as determined by the Gibbs free energy curves, and resulting in different observable volume fractions of each phase. However, at the much slower velocities of the simulation and DLR sample, the low energy pattern dictated by the symmetry of the crystal structure and the resulting interfacial energy, is maintained. At the higher velocities in the UAB sample, local inhomogeneities in the melt produced by convection overwhelm this effect. This is consistent with other recent findings by the Nestler group, showing that pattern selection in these samples is influenced strongly by small changes in the liquid composition. This finding highlights the importance of microgravity experiments. Further analysis of these issues is ongoing. The excellent tomography results of the eutectic structure obtained show the possibility for 3D reconstruction and analysis via synchrotron radiation. Possibilities for tomographic analysis of additional samples will be investigated by the PI in the coming year, as personnel changes at DLR mean that they have withdrawn from the SETA project and will no longer be contributing.
Bibliography Type:	Description: (Last Updated: 06/05/2023)
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